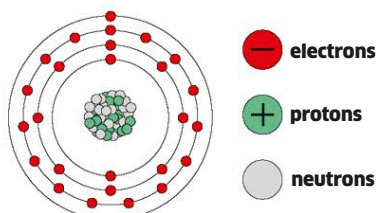


Periodic table key

- Alkali metals
- Alkaline earth metals
- Transition metals
- Lanthanide metals
- Actinide metals
- Other metals
- Metalloids
- Other non-metals
- Halogens
- Noble gases

Key to atomic models on pp.30–41.



Models show atoms of the most common isotope.

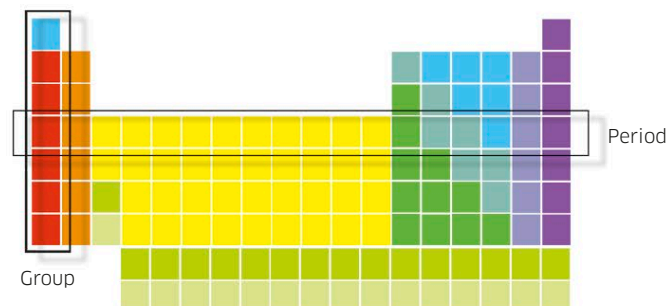
Other non-metals

These include three elements essential for life on Earth—carbon, nitrogen, and oxygen.

					18 2 4.0026 He HELIUM
13 5 10.811 B BORON	14 6 12.011 C CARBON	15 7 14.007 N NITROGEN	16 8 15.999 O OXYGEN	17 9 18.998 F FLUORINE	10 20.180 Ne NEON
13 13 26.982 Al ALUMINUM	14 14 28.086 Si SILICON	15 15 30.974 P PHOSPHORUS	16 16 32.065 S SULFUR	17 17 35.453 Cl CHLORINE	18 18 39.948 Ar ARGON
31 31 69.723 Ga GALLIUM	32 32 72.64 Ge GERMANIUM	33 33 74.922 As ARSENIC	34 34 78.96 Se SELENIUM	35 35 79.904 Br BROMINE	36 36 83.80 Kr KRYPTON
49 49 114.82 In INDIUM	50 50 118.71 Sn TIN	51 51 121.76 Sb ANTIMONY	52 52 127.60 Te TELLURIUM	53 53 126.90 I IODINE	54 54 131.29 Xe XENON
81 81 204.38 Tl THALLIUM	82 82 207.2 Pb LEAD	83 83 208.96 Bi BISMUTH	84 (209) Po POLONIUM	85 (210) At ASTATINE	86 (222) Rn RADON
113 (284) Nh NIHONIUM	114 (289) Fl FLEROVIUM	115 (288) Mc MOSCOVIUM	116 (293) Lv LIVERMORIUM	117 (294) Ts TENNESSINE	118 (294) Og OGANESSON
66 162.50 Dy DYSPROSIUM	67 164.93 Ho HOLMIUM	68 167.26 Er ERBIUM	69 168.93 Tm THULIUM	70 173.04 Yb YTTERBIUM	71 174.97 Lu LUTETIUM
98 (251) Cf CALIFORNIUM	99 (252) Es EINSTEINIUM	100 (257) Fm FERMIUM	101 (258) Md MENDELEVIUM	102 (259) No NOBELIUM	103 (262) Lr LAWRENCIUM

UNDERSTANDING THE PERIODIC TABLE

Within the table are blocks of elements that behave in similar ways. On the left are the most reactive metals. Most everyday metals occur in the middle of the table in a set called the transition metals. Non-metals are mostly on the right of the table and include both solids and gases.

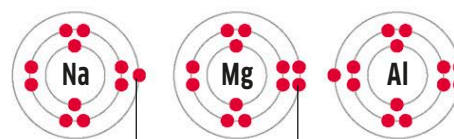


Building blocks

The periodic table is made up of rows called periods and columns called groups. As we move across each period, the elements change from solid metals (on the left) to gases (on the right).

Periods

All elements in a period have the same number of electron shells in their atoms. For example, all elements in the third period have three shells (but a different number of electrons).



One electron in outer shell.

Two electrons in outer shell.

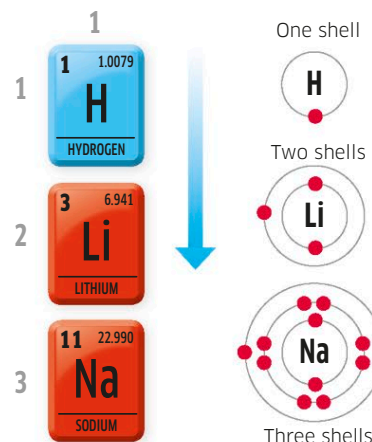
Three electrons in outer shell.

Shrinking atoms

As you move along each row (period) of the table, the atoms of each element contain more protons and electrons. Each atom has the same number of electron shells, but for each step to the right, there are more positively charged protons pulling the shells inward. This "shrinks" the atom, and makes it more tightly packed.

Groups

The elements in a group react in similar ways because they have the same number of electrons in their outer shell (see p.13). For example, while the elements in group 1 all have different numbers of electrons, and shells, they all have just one electron in their outer shell.



Growing atoms

Atoms get bigger and heavier as we move down each column (group). This is because the atoms of each element below have more protons and more electrons than the element above. As shells fill up with electrons (see p.13), a new shell is added each time we move another step down a group, down to the next period.